



OPEN The impact of family cancer history on tumor metabolism and prognosis in patients with non-small cell lung cancer

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Non-small cell lung cancer (NSCLC) is a leading cause of cancer-related mortality, and the role of family cancer history in disease progression and treatment response remains underexplored. This study aims to investigate the influence of family tumor history on disease-free survival, tumor metabolism, and treatment response in NSCLC patients. A retrospective, single-center study of 414 NSCLC patients was conducted, with 101 patients having a family history of cancer (FHC). Disease-free survival (DFS), tumor glucose metabolism assessed by ¹⁸F-FDG PET/CT, and treatment response to chemoradiotherapy, targeted therapy, and immunotherapy were analyzed. Multivariate modeling was performed to improve prognostic prediction. Patients with FHC exhibited higher TNM staging, increased susceptibility to lymph node invasion, and elevated tumor glucose metabolism levels. Family history of cancer, particularly colorectal and lung cancer, was a significant risk factor for disease-free survival. Targeted therapy and immunotherapy significantly improved patient prognosis, while family history of cancer affected the efficacy of chemoradiotherapy but not targeted therapy or immunotherapy. Multivariate modeling combining FHC, treatment, and tumor metabolism levels yielded improved predictive performance. Our study highlights the importance of considering a patient's family history when assessing risk profiles and formulating treatment decisions for NSCLC patients. Further research is needed to understand the molecular mechanisms underlying these observed associations and to develop more effective treatment strategies for NSCLC patients with a cancer family history.

Keywords NSCLC, Prognosis, Disease-free survival (DFS), ¹⁸F-FDG PET/CT, Family history of cancer (FHC)

Lung cancer remains a leading cause of cancer-related mortality worldwide, with NSCLC accounting for approximately 85% of all cases, and it is associated with high morbidity and mortality rates worldwide^{1,2}. Despite advancements in diagnostic and therapeutic approaches, disease-free survival rates have not substantially improved, underscoring the need to identify novel prognostic factors and optimize treatment strategies. One potential avenue for enhancing our understanding of lung cancer progression and response to therapy is the investigation of family tumor history. Family history has long been recognized as a risk factor for the development of cancer, with a higher likelihood of occurrence in individuals with first-degree relatives diagnosed with the disease^{3,4}. This increased risk is likely attributed to a combination of genetic predisposition, shared environmental factors, and lifestyle habits passed down through generations^{5,6}. However, the relationship between family tumor history and disease-free survival in lung cancer patients remains less well characterized.

In this study, we aim to systematically explore the influence of family tumor history on disease-free survival in a cohort of lung cancer patients. Specifically, we focus on the impact of parental and sibling history, tumor type in the family, and glucose metabolism levels assessed by ¹⁸F-fluorodeoxyglucose(¹⁸F-FDG) positron emission tomography/computed tomography (PET/CT). Furthermore, we analyze the interplay between family tumor

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	Total <i>n</i> = 414	Without <i>n</i> = 313	With <i>n</i> = 101	<i>P</i> value
Age	62.0 [55.0;69.0]	62.0 [55.0;68.0]	63.0 [54.0;70.0]	0.884
Gender:				0.252
Female	119 (28.7%)	95 (30.4%)	24 (23.8%)	
Male	295 (71.3%)	218 (69.6%)	77 (76.2%)	
Family history type of cancer:				<0.001
None	313 (75.6%)	313 (100%)	0 (0.00%)	
Colon cancer	18 (4.35%)	0 (0.00%)	18 (17.8%)	
Liver cancer	12 (2.90%)	0 (0.00%)	12 (11.9%)	
Lung cancer	34 (8.21%)	0 (0.00%)	34 (33.7%)	
Others	37 (8.94%)	0 (0.00%)	37 (36.6%)	
Family history family category:				<0.001
None	313 (75.6%)	313 (100%)	0 (0.00%)	
Parent	42 (10.1%)	0 (0.00%)	42 (41.6%)	
Parent and Sibling	21 (5.07%)	0 (0.00%)	21 (20.8%)	
Sibling	38 (9.18%)	0 (0.00%)	38 (37.6%)	
Treatment:				0.707
Chemoradiotherapy	247 (59.7%)	184 (58.8%)	63 (62.4%)	
Immunotherapy	64 (15.5%)	48 (15.3%)	16 (15.8%)	
Targeted therapy	103 (24.9%)	81 (25.9%)	22 (21.8%)	
TNM stage:				0.005
IB	144 (34.8%)	121 (38.7%)	23 (22.8%)	
II	173 (41.8%)	120 (38.3%)	53 (52.5%)	
IIIA	97 (23.4%)	72 (23.0%)	25 (24.8%)	
Lymphnode invasion:				0.022
Without	301 (72.7%)	237 (75.7%)	64 (63.4%)	
With	113 (27.3%)	76 (24.3%)	37 (36.6%)	
Differentiation:				0.124
Poor	97 (23.4%)	70 (22.4%)	27 (26.7%)	
Moderate	207 (50.0%)	152 (48.6%)	55 (54.5%)	
Well	110 (26.6%)	91 (29.1%)	19 (18.8%)	
Pathology:				0.114
Adenocarcinoma	259 (62.6%)	203 (64.9%)	56 (55.4%)	
Squamous carcinoma	155 (37.4%)	110 (35.1%)	45 (44.6%)	
Size	3.20 [2.20;4.50]	3.10 [2.10;4.50]	3.50 [2.60;4.70]	0.060
SUVmax	6.80 [3.50;10.5]	6.10 [3.00;9.19]	9.10 [6.00;12.5]	<0.001

Table 1. Correlation between family history of cancer and clinicopathologic parameters.

history and various treatment modalities, including chemoradiotherapy, targeted therapy, and immunotherapy, to better understand their combined effects on patient outcomes.

Methods

Study population

In this retrospective, single-center study, we employed the following inclusion and exclusion criteria to screen patients, inclusion criteria: Individuals newly diagnosed who have not previously received any related treatment, 18 years of age or older, with non-small cell lung cancer confirmed through pathology (clinical stage IB-III according to AJCC)⁷; Complete clinical and biochemical data available; Underwent surgical treatment between 2013 and 2020; Family history information obtainable; Postoperative treatment is clear; No presence of other malignancies or serious ailments; An ¹⁸F-FDG PET/CT scan was conducted within 45 days prior to the surgery. Exclusion criteria: Inability to obtain complete clinical and biochemical data; The tumor has metastasized to distant sites; Presence of other malignant tumors or other serious diseases; Incomplete or unreliable family history information.

Treatment was determined based on whether patients received chemoradiotherapy alone or in combination with immunotherapy or targeted therapy in the first year after surgery. Immunotherapy drugs include but are not limited to Nivolumab, Sinitilimab, etc. Targeted therapeutic drugs include but are not limited to Gefitinib, Bevacizumab, etc. The outcome of this study is disease-free survival, which is defined as the time from the date

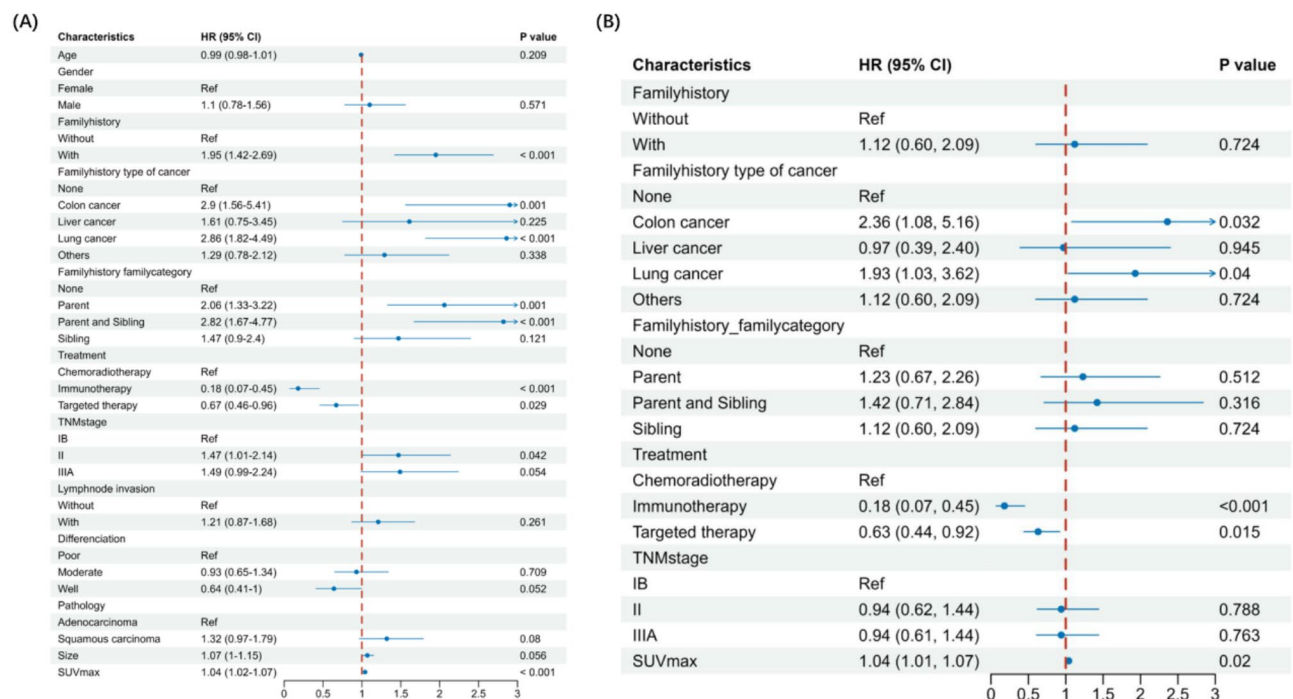


Fig. 1. The hazard ratio of DFS among each subgroup shown in the forest plot. Univariate (A) and multivariate (B) regression analysis. CI, confidence interval.

of surgery to the recurrence of the tumor, tumor-related death, or the last visit date. By the follow-up deadline of July 30th, 2023, a total of 414 patients were incorporated into the study.

The protocol of our study was reviewed and approved by the ethics committee of The Hunan Cancer Hospital (No.23316), and the methods were carried out in accordance with the principles of the Declaration of Helsinki. Informed consent was obtained from all subjects at follow-up.

Assessment of family tumor history

We defined a family history of cancer as having at least one first-degree relative diagnosed with cancer, the first-degree relative included parents and siblings⁸. The type of cancer is also documented, we aim to further analyze the impact of family members' tumor types on NSCLC patients, patients with more than two first-degree relatives suffering from tumors with different tumor types were not included in the study.

Evaluation of glucose metabolism using PET/CT

In our study, patients were instructed to fast for at least 6 h before undergoing ¹⁸F-FDG PET/CT scans. PET/CT images were reviewed and analyzed by experienced nuclear medicine physicians using dedicated software. Regions of interest (ROIs) were drawn around the primary lung tumors, and the maximum standardized uptake value (SUVmax) was recorded for each tumor. The SUVmax represents the highest glucose metabolism level within the tumor and serves as a semi-quantitative measure of tumor glycolytic activity. The measurement of lesion SUVmax is consistent between different observers; however, potential influencing factors include environmental temperature, physical activity, blood glucose levels, medications, hydration, body mass index, and the timing of the measurement. The imaging protocol in this study was strictly performed according to the "FDG PET/CT: EANM procedure guidelines for tumour imaging: version 2.0"⁹, which minimized the interference of these factors on SUVmax to the greatest extent possible.

Statistical analysis

To examine differences across groups, for continuous variables that meet the conditions of normal distribution and homogeneity of variance, we use the independent samples t-test for intergroup comparisons. For continuous variables that do not meet the conditions of normal distribution or homogeneity of variance, we use the Mann-Whitney U test for intergroup comparisons, and the chi-square test or Fisher exact test for categorical variables. Survival curves were drawn by the Kaplan-Meier method, and the log-rank test was performed to examine survival variations among groups. Univariate analysis of possible risk variables is conducted using the Cox proportional hazard model. In the multivariate COX regression analysis, the factors having a P value of under 0.05 in the univariate COX analysis were included, the hazard ratios (HRs) along with their corresponding 95% confidence intervals (CIs) are provided to assess the magnitude and precision of the effect estimates. Time-dependent AUC is used to quantify the distinguishing performance of features. R program (version 4.12) was used for all statistical analyses. Statistically significant are those with bilateral p values less than 0.05.

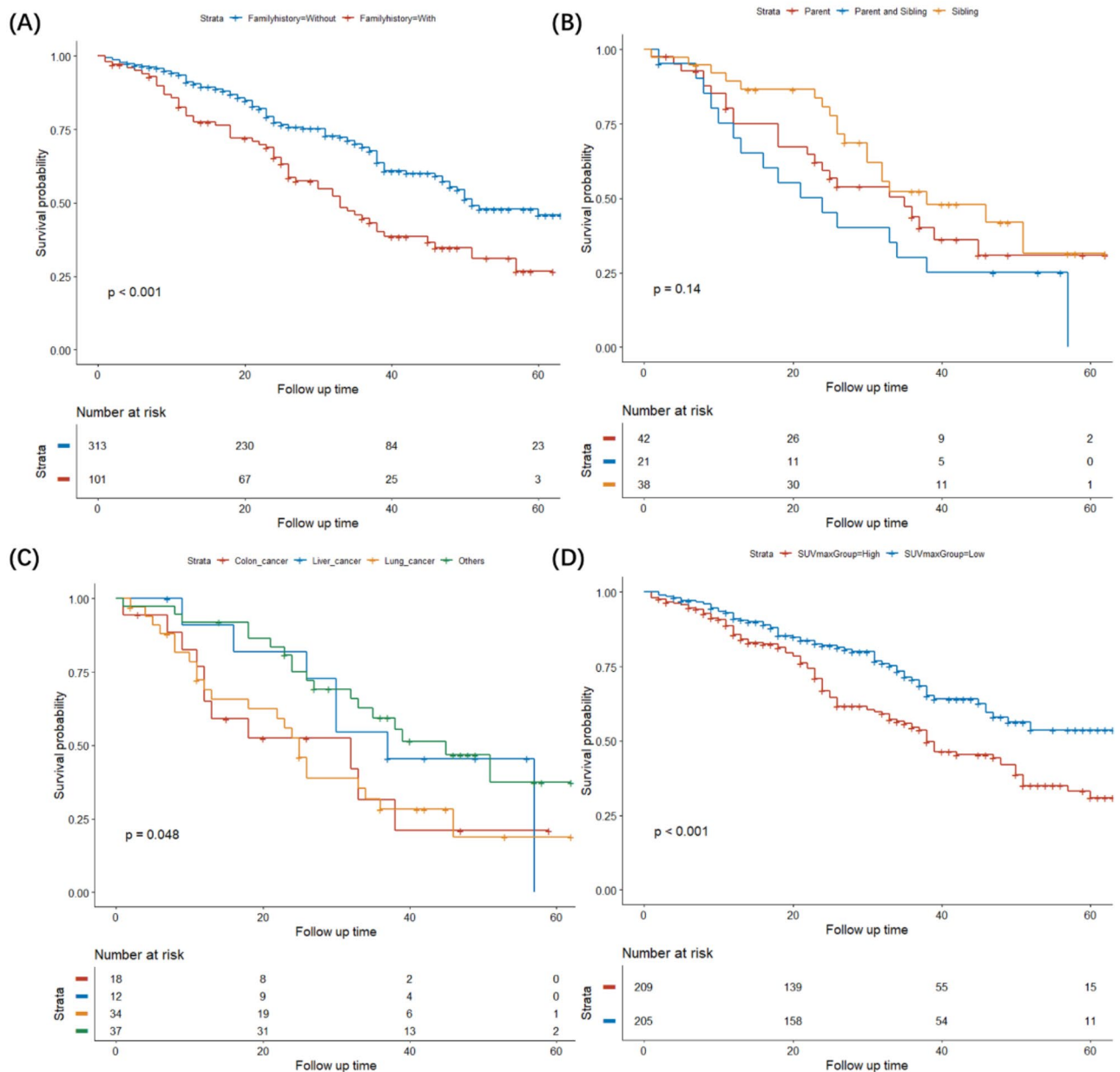


Fig. 2. Kaplan-Meier (KM) survival curves for non-small cell lung carcinoma patients. **(A)** DFS stratified by familial history of cancer. **(B)** DFS stratified by the incidence of cancer in parents and siblings. **(C)** DFS stratified by type of tumor in first-degree relative. **(D)** DFS stratified by SUVmax.

Results

Patient characteristics

414 individuals were enrolled in this study, of which 101 had a familial history of cancer and 313 lacked such history. No significant differences in age or gender were observed between the two groups. Likewise, tumor differentiation levels and pathological types revealed no discrepancies. However, patients with FHC exhibited higher TNM staging and larger sizes, were more susceptible to lymph node invasion, and demonstrated elevated tumor glucose metabolism levels (Table 1).

Familial history of cancer impacts the disease-free survival for NSCLC patients

Based on the results of our risk factor forest plot, univariate analysis revealed that having a family history of cancer is a significant risk factor, with colorectal and lung cancer family histories having a more pronounced impact on prognosis. Regarding treatment modalities, targeted therapy and immunotherapy both considerably improved prognoses. Neither pathological differentiation degree nor pathological type affected the prognosis, while a high SUVmax of tumor lesions emerged as a risk factor. Multivariate analysis results indicated that family history of cancer type, immunotherapy, targeted therapy, and tumor SUVmax influenced patient outcomes

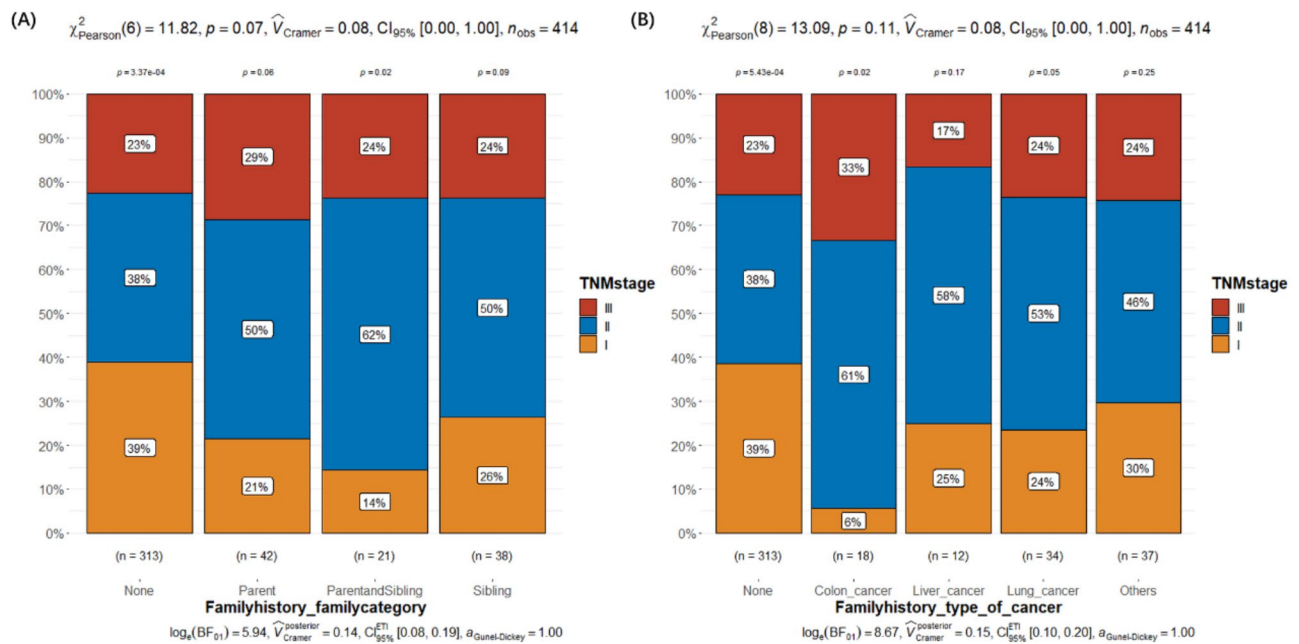


Fig. 3. (A) The staging of TNM in lung cancer patients with different family tumor history, (B) based on the type of cancer in the family history.

(Fig. 1). The KM survival curve further illustrates this result (Fig. 2). We analyzed male and female patients separately, and the results showed that a family history of cancer affects both male and female patients (figure S1). We further analyzed the TNM staging composition of patients with FHC, and found no differences in TNM composition between groups, irrespective of whether they were classified by family category or family history type of cancer (Fig. 3).

Family history of cancer is associated with tumor glucose metabolism in NSCLC patients

We analyzed the association between familial history of cancer and patients' tumor glucose metabolism levels. Our findings indicated that NSCLC patients with a family history of neoplasms exhibited higher glucose metabolism levels in primary lesions. When both parents and siblings had cancer, the tumor SUVmax was higher. Further categorization based on the type of cancer in the family history showed that patients with a lung cancer family history also presented higher SUVmax values (Fig. 4). This suggests that a family history of cancer may impact the glucose metabolism levels of primary lesions in NSCLC patients.

Family history of cancer and treatment

Differences in treatment modalities can impact patient prognosis. We further examined the influence of familial cancer history on patient survival under different treatment. Our findings revealed that immunotherapy and targeted therapy significantly improved patient prognosis, while the FHC did not affect the efficacy of these treatments. For patients receiving chemoradiotherapy alone, those with a family history of neoplasms had a notably reduced survival period. Our results suggest that FHC may affect the efficacy of chemoradiotherapy but may not impact the effectiveness of targeted therapy or immunotherapy (Fig. 5).

Multivariate modeling provides superior prognostic prediction for patients

We constructed ROC curves to analyze the prognostic prediction efficacy of various parameters. Certain individual indicators demonstrated high predictive performance in the first year, such as family history tumor type (AUC = 0.753) and treatment method (AUC = 0.700). However, their predictive performance for three- and five-year periods was relatively low. Combining FHC, treatment and tumor metabolism levels in a multivariate model yielded improved predictive performance at one year (AUC = 0.701), three years (AUC = 0.661), and five years (AUC = 0.683). Consequently, FHC may play a crucial role in the prognosis assessment of patients (Fig. 6).

Discussion

The present study provides valuable insights into the influence of a family history of cancer on tumor metabolism and disease-free survival in NSCLC patients undergoing different treatment modalities. Our findings have important implications for understanding the biological mechanisms underlying NSCLC progression and informing clinical decision-making for these patients.

Our results demonstrate that patients with FHC exhibited higher TNM staging, increased susceptibility to lymph node invasion, and elevated tumor glucose metabolism levels. This finding aligns with earlier research, which has reported that genetic predisposition may contribute to more aggressive tumor characteristics in

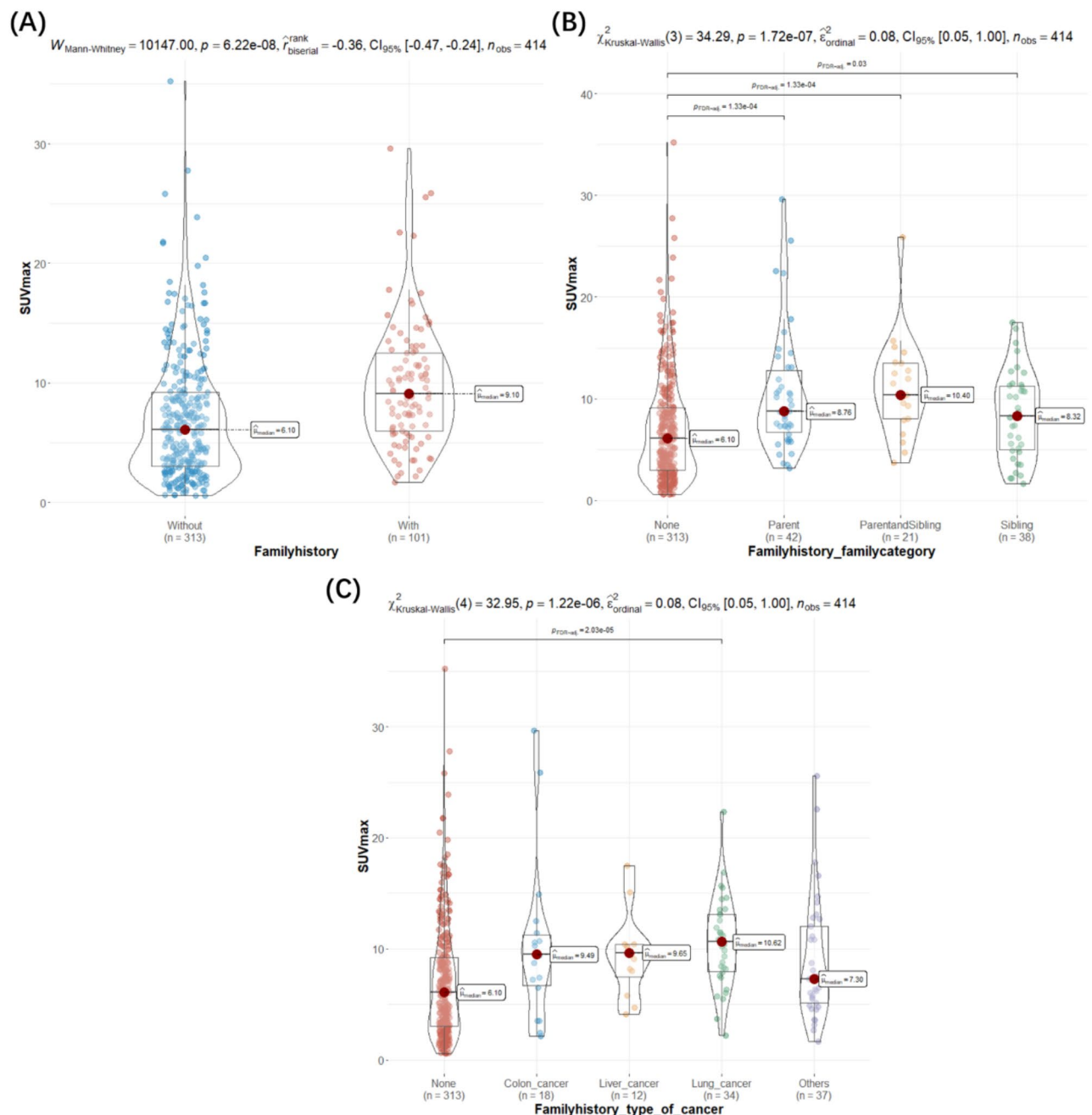


Fig. 4. Tumor glucose metabolism level (SUVmax). **(A)** Stratified by familial history of cancer. **(B)** Stratified by the incidence of cancer in parents and siblings. **(C)** Stratified by type of tumor in first-degree relative.

various cancer types^{10,11}. Additionally, relevant studies have mentioned that NSCLC patients with a cancer family history are more likely to exhibit epidermal growth factor receptor (EGFR) mutations¹². Glucose metabolism, as a vital metabolic process in tumor, influences tumor initiation and progression. Genetic traits are closely associated with tumor glucose metabolism, making the relationship between family history and tumor glucose metabolism particularly intriguing^{13,14}. This connection may provide new insights into the genetic factors driving tumor metabolism in NSCLC patients. Further research is needed to elucidate the specific molecular mechanisms that link family history to tumor metabolism.

The impact of family history on disease-free survival in our study is in agreement with previous research emphasizing the role of genetic factors in cancer prognosis¹⁵. Our findings suggest that having a family history of cancer, particularly colorectal and lung cancer, is a significant risk factor for disease-free survival in NSCLC patients. Numerous studies have mentioned the interplay and regulation between the lungs and the gastrointestinal tract^{16,17}. The oncogenic KRAS gene, commonly mutated in both lung and colorectal cancers^{18–21}, is influenced by hereditary traits²², which could potentially explain the influence of colorectal cancer family history on the

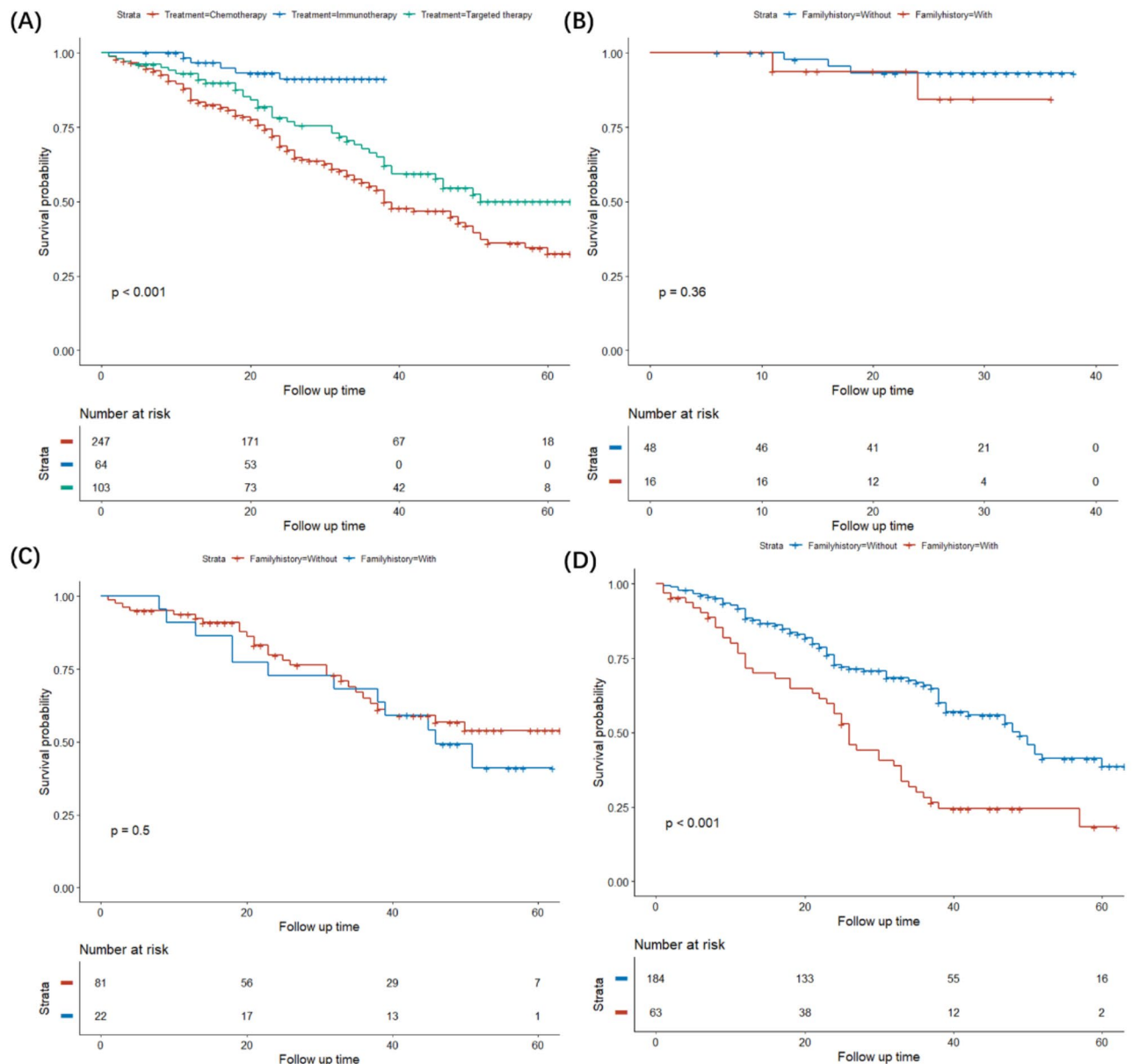


Fig. 5. The effect of family tumor history on DFS in patients with NSCLC under different treatment. (A) KM curves of patients receiving different treatments, (B) KM curve of patients receiving immunotherapy with or without family history of tumor. (C) KM curve of patients receiving targeted therapy with or without family history of tumor. (D) KM curve of patients receiving chemoradiotherapy with or without family history of tumor.

survival of NSCLC patients. Additionally, we observed that high SUVmax of tumor lesions is a risk factor for poor prognosis, which is consistent with prior studies reporting a correlation between high glucose metabolism and poor survival in various cancer types^{23–25}.

With the application of targeted therapy and immunotherapy, lung cancer patients have access to improved treatment options, and immunotherapy has demonstrated significant efficacy in certain cases. Our research indicates that both immunotherapy and targeted therapy can significantly enhance patient prognosis. This finding supports previous studies that have reported the benefits of these treatments in NSCLC patients^{26–28}. The presence or absence of a cancer family history does not impact the efficacy of targeted therapy or immunotherapy, with no differences in disease-free survival. Concurrently, we discovered that a cancer family history may affect the effectiveness of chemoradiotherapy, as patients with FHC receiving only chemoradiotherapy exhibited reduced survival periods. Certain genetic mutations, such as BRCA1/2, can affect the sensitivity of tumors to specific chemotherapy drugs, KRAS and BRAF mutations are often associated with poorer disease-free survival (DFS)²⁹. These mutations may be present in patients with a family history of cancer, potentially affecting the efficacy of radiotherapy and chemotherapy in these individuals. In contrast, targeted therapy and immunotherapy depend

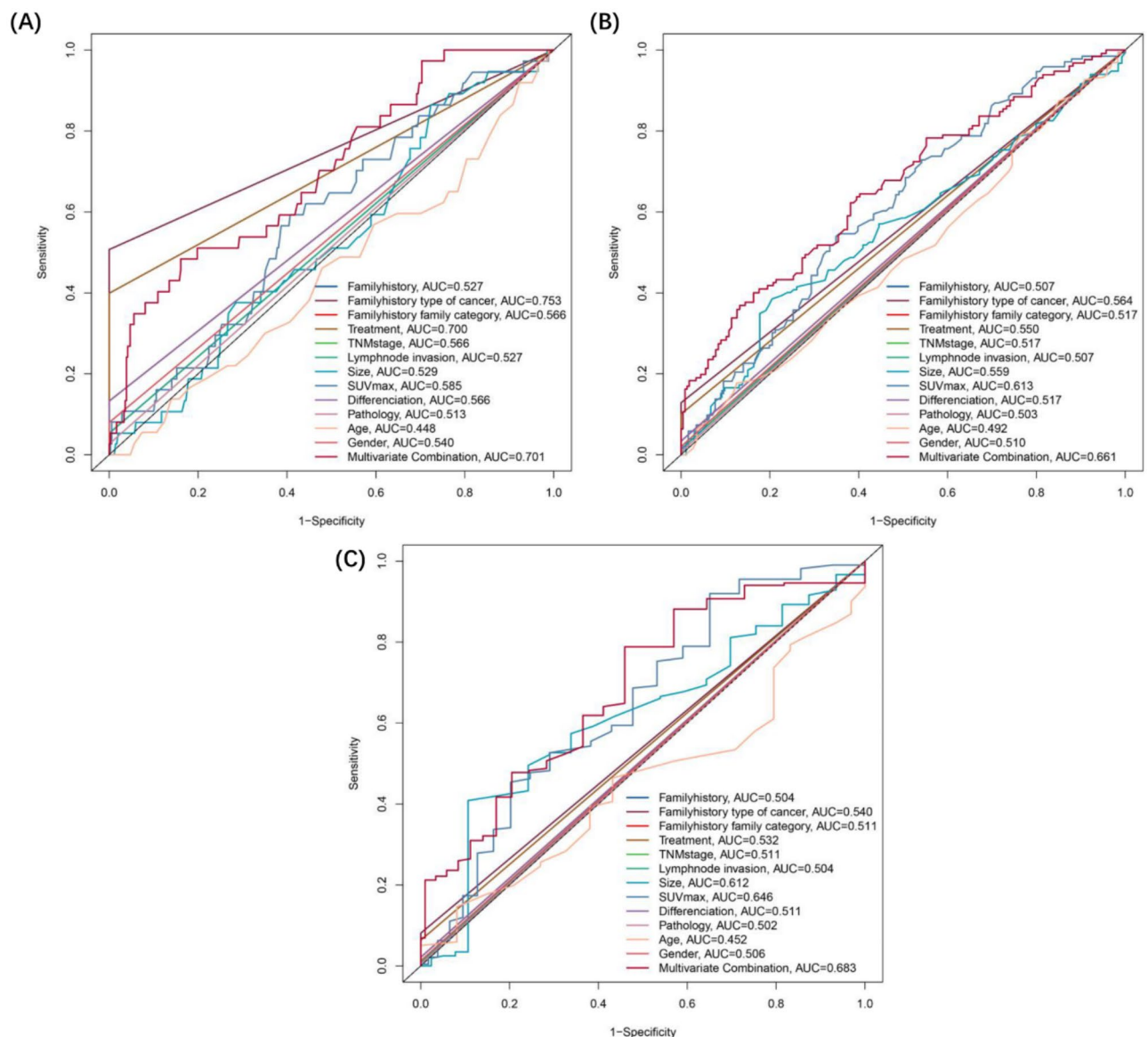


Fig. 6. ROC analysis of various indexes on DFS. (A) 1 year, (B) 3 years, (C) 5 years. Multivariate combination: all meaningful items in the multivariate regression analysis.

on specific molecular targets and the immune environment, which may not be related to genetic factors but rather rely more on the intrinsic characteristics and microenvironment of the tumor³⁰. This observation suggests that genetic factors related to family history may influence tumor response to chemoradiotherapy, necessitating further research to optimize treatment strategies for such patients.

Our study has several limitations, including its retrospective nature and relatively small sample size, which may introduce biases to the results. Further research is needed to confirm and expand upon our findings. Since we aim to analyze the impact of tumor types in family members on NSCLC patients, we excluded patients with more than two first-degree relatives diagnosed with different types of tumors. Consequently, some patients with multiple affected relatives were not included in the study. The presence of multiple affected relatives or just one affected relative might also influence the prognosis of NSCLC patients. We will explore this aspect in future research. Investigating the potential mechanisms by which genetic factors influence lung cancer progression is also crucial, in order to develop more effective treatments for patients with a cancer family history.

In conclusion, our study emphasizes the importance of considering a patient's family history when assessing risk profiles and formulating treatment decisions for non-small cell lung carcinoma patients. Future research should aim to identify the potential molecular mechanisms underlying the observed associations between family history, tumor metabolism, and treatment response, which may ultimately provide more effective treatment strategies for non-small cell lung carcinoma patients with a cancer family history.

Data availability

The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request.

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Author contributions

Mengtian Ma and Hongpei Tan designed and conceived the project. Haixiong Yan contributed to the statistics. Kai Zheng contributed to the design of this work, image interpretation, and revised the important key content of the manuscript. All authors reviewed the manuscript and approved the submitted version.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics statement

The protocol of our study was reviewed and approved by the ethics committee of The Hunan Cancer Hospital (No.23316), and the methods were carried out in accordance with the principles of the Declaration of Helsinki. Informed consent was obtained from all subjects at follow-up.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-024-73080-w>.

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